Name: Student ID:

Please write your name clearly in UPPERCASE

MAST20030 Differential Equations mid-semester test

Exam Instructions:

- Write your name and student ID in the boxes at the top of this page. You can do this immediately.
- Do not open this booklet until instructed to do so by the exam supervisors.
- Use the coloured paper for rough working. Anything you write on the coloured paper will not be marked. You can request additional coloured paper for rough working by raising your hand.
- Write your answers on this test paper in the spaces provided.
- If you need more paper for answering a question, raise your hand, and request white paper. Write your student ID at the top of the additional page.
- You can write with a pen or a pencil.
- You cannot leave early. If you finish early, check your work.
- Place your student ID card face-up on the desk.
- You must use methods taught in MAST20030 Differential Equations to solve the questions.
- Show full working in your solutions, unless the question states otherwise.

Exam conditions:

- There are 45 minutes writing time. There is no reading time.
- The test consists of 3 questions on 3 pages (not including this cover page).
- No other materials are permitted.
- Calculators are not permitted.
- Communicating with others is not permitted.
- This test contributes 15% of your MAST20030 grade.

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Question 1. (6 marks)
Write the following ODE systems in vector–matrix form, $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y}$, and calculate the eigenvalues of the matrices $\boldsymbol{A}$.

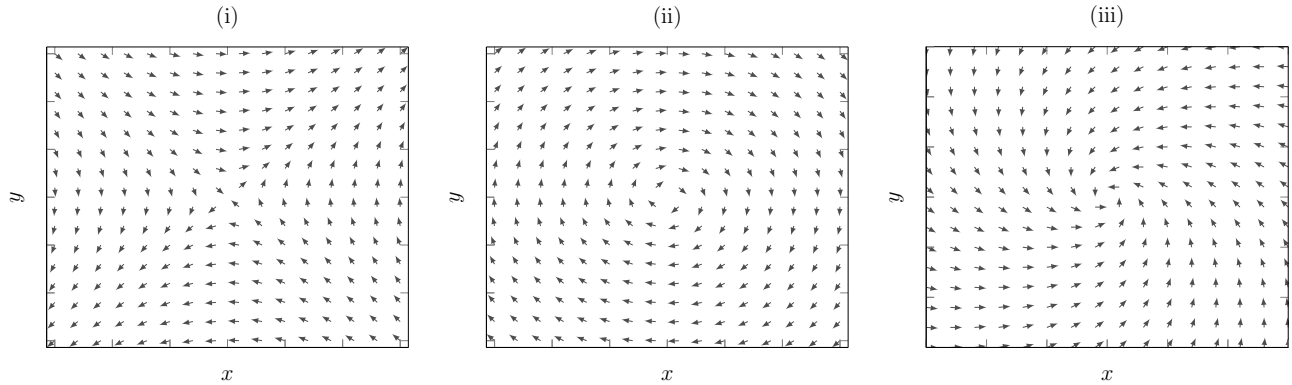
$$\left.\begin{aligned} x' &= y \\ y' &= -x \end{aligned}\right\} (a) \qquad \left.\begin{aligned} x' &= -x - y \\ y' &= x - y \end{aligned}\right\} (b) \qquad \left.\begin{aligned} x' &= y \\ y' &= x \end{aligned}\right\} (c)$$

$$(a)\ A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}; \quad (b)\ A = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}; \quad (c)\ A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The eigenvalues are

$$(a)\ \lambda = \pm \mathrm{i}; \quad (b)\ \lambda = -1 \pm \mathrm{i}; \quad (c)\ \lambda = \pm 1.$$

Match the systems (a–c) with the below direction fields. Justify your answers based on the eigenvalues of the systems.



Based on the eigenvalues, the system (a) is a stable centre, (b) is an asymptotically stable spiral node, and (c) is an unstable saddle. Therefore, they correspond to the direction fields as:

$$(a) \leftrightarrow (ii) \quad (b) \leftrightarrow (iii) \quad (c) \leftrightarrow (i).$$

Question 2. (5 marks)
Find the general solution of the following ODE using a power series about $x = 0$,

$$(1 - x^2)\, y''(x) - x\, y'(x) + y(x) = 0.$$

Find the radius of convergence of any infinite series involved in the solution.

Use a power series of form

$$y(x) = \sum_{n=0}^{\infty} c_n\, x^n.$$

Substituting into the ODE gives

$$\sum_{n=0}^{\infty} \left\{ (1 - x^2)\, n\, (n-1)\, c_n\, x^{n-2} - x\, n\, c_n\, x^{n-1} + c_n\, x^n \right\} = 0$$

$$\Rightarrow \quad \sum_{n=0}^{\infty} \left\{ c_n\, n\, (n-1)\, x^{n-2} + c_n\, [1 - n - n\,(n-1)]\, x^n \right\} = 0$$

$$\Rightarrow \quad \sum_{m=0}^{\infty} \left\{ c_{m+2}\, (m+2)\, (m+1) + c_m\, (1 - m^2) \right\} x^m = 0$$

$$\Rightarrow \quad c_{m+2} = \frac{m^2 - 1}{(m+2)\,(m+1)} c_m = \frac{m-1}{m+2} c_m \quad \text{for} \quad m = 0, 1, 2, \ldots$$

Let $c_0 = 1$ and $c_1 = 1$, then

$$c_2 = \frac{-1}{2}, \quad c_4 = \frac{c_2}{4} = \frac{-1}{8}, \quad \text{and so on,}$$

and $c_3 = c_5 = \ldots = 0$.
Two linearly independent solutions are

$$y_1(x) = \sum_{n=0}^{\infty} c_{2\,n} x^{2\,n} \quad \text{and} \quad y_2(x) = x \quad \Rightarrow \quad y(x) = C_1\, y_1(x) + C_2\, y_2(x)$$

is the general solution to the ODE, where C_1 and C_2 are arbitrary constants.
Apply the ratio test to y_1

$$\lim_{n\to\infty} \left| \frac{c_{2\,n+2}\, x^{2\,n+2}}{c_{2\,n} x^{2\,n}} \right| = \lim_{n\to\infty} \left| \frac{2\,n - 1}{2\,n + 2} \right| x^2 = x^2,$$

which is < 1 if $|x| < 1$, i.e., the radius of convergence is one.

Question 3. (4 marks)
Find the inverse Laplace transforms

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2\,s+3}{s^2-s}\right\} \quad \text{and} \quad g(t) = \mathcal{L}^{-1}\left\{\frac{\mathrm{e}^{-\pi\,s}}{s^2+1}\right\}.$$

Show your workings, and state any properties or theorems you use.

You may use without proof that

$$\mathcal{L}\{\mathrm{e}^{a\,t}\} = \frac{1}{s-a}; \quad \mathcal{L}\{\sin a\,t\} = \frac{a}{s^2+a^2}; \quad \mathcal{L}\{k(t-a)\,H(t-a)\} = \mathrm{e}^{-a\,s}\,K(s).$$

Use partial fractions to find $f(t)$, as follows

$$\begin{aligned}\frac{2\,s+3}{s^2-s} &= \frac{A}{s} + \frac{B}{s-1}\\ &= \frac{A\,(s-1)+B\,s}{s^2-s}\\ &= \frac{(A+B)\,s-A}{s^2-s}\end{aligned}$$

so that

$$A+B=2, \quad -A=3 \quad \Rightarrow \quad A=-3, \quad B=5.$$

Therefore

$$\begin{aligned}f(t) = \mathcal{L}^{-1}\left\{\frac{2\,s+3}{s^2-s}\right\} &= \mathcal{L}^{-1}\left\{\frac{5}{s-1} - \frac{3}{s}\right\}\\ &= 5\,\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - 3\,\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} \quad \text{by linearity of IFT}\\ &= 5\,\mathrm{e}^{t} - 3.\end{aligned}$$

For $g(t)$, we have

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{\mathrm{e}^{-\pi\,s}}{s^2+1}\right\} &= \mathcal{L}^{-1}\left\{\mathrm{e}^{-\pi\,s}\,A(s)\right\} \quad \text{where } A(s) = 1/(s^2+1)\\ &= a(t-\pi)\,H(t-\pi) \quad \text{by } t\text{-shifting theorem}\end{aligned}$$

Now

$$a(t) = \mathcal{L}^{-1}\{A(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin(t).$$

Therefore

$$g(t) = \mathcal{L}^{-1}\left\{\frac{\mathrm{e}^{-\pi\,s}}{s^2+1}\right\} = \sin(t-\pi)\,H(t-\pi)$$

End of test.